USBCANFD Analyzer Manual

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Zhuhai Chuangxin Technology Co., Ltd
Email：zhcxgd@163.com

Revision History

Version	Date	Description
V1.0	2021.12.27	First Edition
V1.1	2023.06.17	Update Software Screenshots
V1.2	2023.10.29	Panel Information Corrections
V1.3	2024.01.22	Update Software Screenshots
V1.4	2025.06.12	Corrected picture of interface switch

Contents

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1 Equipment Introduction

1.1 Overview

The USBCANFD analyzer is a CAN analyzer with a high-speed USB2.0 interface and a 2-channel CANFD interface, and it has the function of sending and receiving CAN/CANFD bus data.

The USBCANFD analyzer can be used as a standard CAN/CANFD node and is a powerful tool for product development, CAN/CANFD bus device testing and data analysis in CAN bus and especially CANFD application scenarios. With this device, a PC can be connected to a standard CAN/CANFD network through the USB interface, which can be used to build fieldbus test labs, industrial control, intelligent buildings, automotive electronics and other fields for data processing, data acquisition and data communication. At the same time, USBCANFD analyzer has a small size, easy to install and other characteristics, is also the best choice for portable system users.

The device CAN/CANFD bus circuit of the equipment adopts independent isolated DC-DC power supply module and high-speed magnetic coupling isolation module for Galvanic isolation, which makes the interface adapter that has strong anti-interference ability and greatly improves the reliability of the system in harsh environments. The USB, CAN1, and CAN2 terminals are completely isolated.

The product can use the CANFD Tool software provided by the manufacturer to directly configure, send and receive the CAN/CANFD bus. Users can also refer to the provided DLL dynamic connection library and VC routines to write their own application programs, facilitating the development of CAN/CANFD system application software products. When using the USBCANFD analyzer for secondary development, you do not need to understand the complex USB interface communication protocol at all.

1.2 Performance and Technical Indicators

·Direct power supply from the USB bus, no external power supply required;

·Protocol conversion between USB and CAN/CANFD bus;

·The USB interface supports USB3.0, USB2.0, compatible with USB1.1 and OTG specifications;

·With 2 independent CAN/CANFD channels;

·Support CAN2.0A, CAN2.0B, ISO11898-1, and CANFD protocol specification V.1.0, support standard frames and extended frames;

·CAN channel baud rate up to 5 Mbps (CANFD), software configurable; ;

·The CAN frame data domain can reach up to 64 bytes per frame (CANFD);

·The CAN bus interface adopts high-speed magnetic coupling isolation and isolated DC-DC power supply; USB, CAN1, CAN2 completely isolated between the three ends;

·Support bidirectional transmission, CAN/CANFD sending, CAN/CANFD receiving;

·Support data frame and remote frame formats;

·Maximum traffic: Receive 18000+fps/channel, transmit 8000+fps/channel, and both channels can operate independently at the same time without affecting each other;

·Internal CAN receive buffer capacity: 1500+frames/channel; CAN transmission hardware buffer capacity: 64 frames/channel (automatic retransmission in case of failure);

·Insulation voltage of isolation module: 2500V;

·Operating Temperature: -40 ℃~105 ℃;

·Shell size: 93 * 69.7 * 24.3mm;

·Product Compatibility: The function library interface is compatible with the Zhou Ligong USBCANFD-200U library interface.

1.3 Typical Applications

·Transmission and reception of CAN/CANFD bus network through USB interface of PC or laptop;

·Fast CAN/CANFD network data collection and analysis;

·CAN/CANFD bus - USB gateway；

·USB interface to CAN/CANFD network interface;

·Extension of CAN/CANFD bus network communication length;

·Industrial Field CAN/CANFD network data monitoring.

1.4 Sales List

Table 1. USBCANFD Analyzer Sales List

Num	Name	Number	Unit	Description
1				
2				
3				
4				
5				
6				

1.5 Technical Support and Services

No reason to return or exchange in 7 days, 5 years free maintenance, lifetime maintenance and upgrade service.

For technical support and purchase information, please refer to: https://www.cxcan.com/

Email: zhcxgd@163.com

Technical Support QQ：3259558860

2 Equipment Appearance and interface

2.1 Equipment Appearance

Figure 1: Front and Back of the Equipment

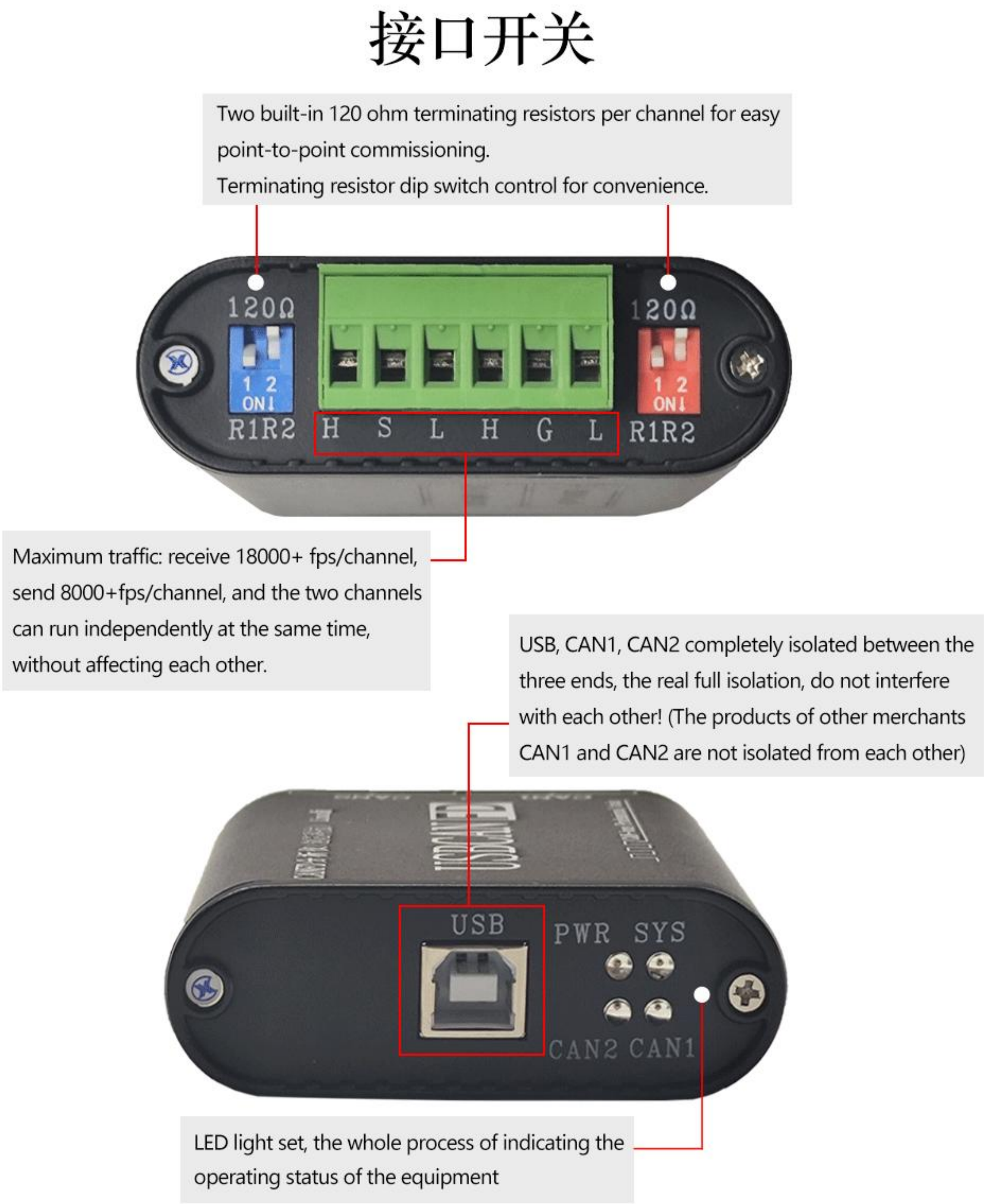


Figure 2: Equipment Side

2.2 Interface Definition

The CANFD analyzer has two sets of external interfaces, distributed on the front and rear panels.

2.2.1 Front Panel

The schematic diagram of the front panel interface is shown in Figure 3, which provides a USB interface and LED light group.

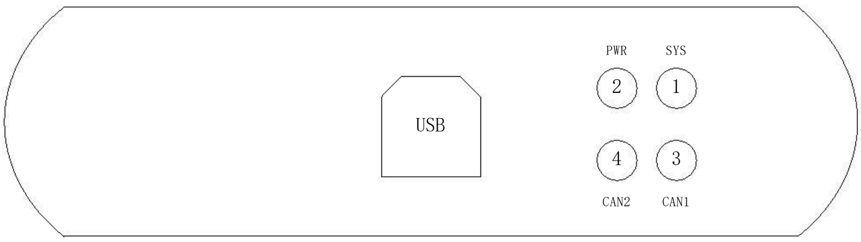


Figure 3: Schematic diagram of the front panel interface of the USBCANFD analyzer

Table 2. Definition of USBCANFD analyzer front panel signals

Num	Name	Definition
1	USB	Device USB interface.
2	SYS	System status indication, normally in a turn off state. Illuminates when there is an error on the bus。
3	PWR	Power indicator light. Always red.
4	CAN1	CAN1 Channel indicator light. Blinking when there is data.
5	CAN2	CAN2 Channel indicator light. Blinking when there is data.

2.2.2 Rear Panel

The rear panel interface is shown in Figure 4, and the detailed definition is shown in Table 3.

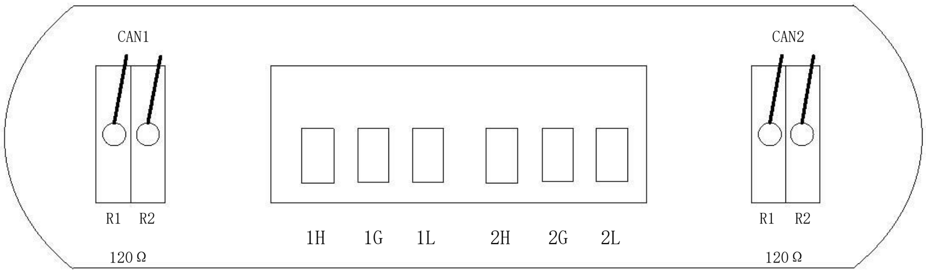


Figure 4 Schematic diagram of the rear panel interface of the USBCANFD analyzer

Table 3. Definition of USBCANFD Analyzer Rear Panel Interface

Num	Name	Description
1	CAN1/R1	CAN1 terminal resistor R1. If turned down to the ON state, the internal 120 ohm resistor will

		be connected to the bus
2	CAN1/R2	CAN1 terminal resistor R2. Parallel with R1, with the same effect. Two built-in resistors per channel
3	1H	CAN1 channel CAN bus H signal
4	1S	CAN1 channel shielded wire interface, if the communication line is shielded, it can be connected to the shielding layer, otherwise it can be grounded or not connected
5	1L	CAN1 channel CAN bus L signal
6	2H	CAN2 channel CAN bus H signal
7	2G	CAN2 channel shielded wire interface, if the communication line is shielded, it can be connected to the shielding layer, otherwise it can be grounded or not connected
8	2L	CAN2 channel CAN bus L signal
9	CAN2/R1	CAN2 terminal resistor R1. If turned down to the ON state, the internal 120 ohm resistor will be connected to the bus
10	CAN2/R2	CAN2 terminal resistor R2. Parallel with R1, with the same effect. Two built-in resistors per channel

3 Instructions for Use

3.1 Factory Configuration

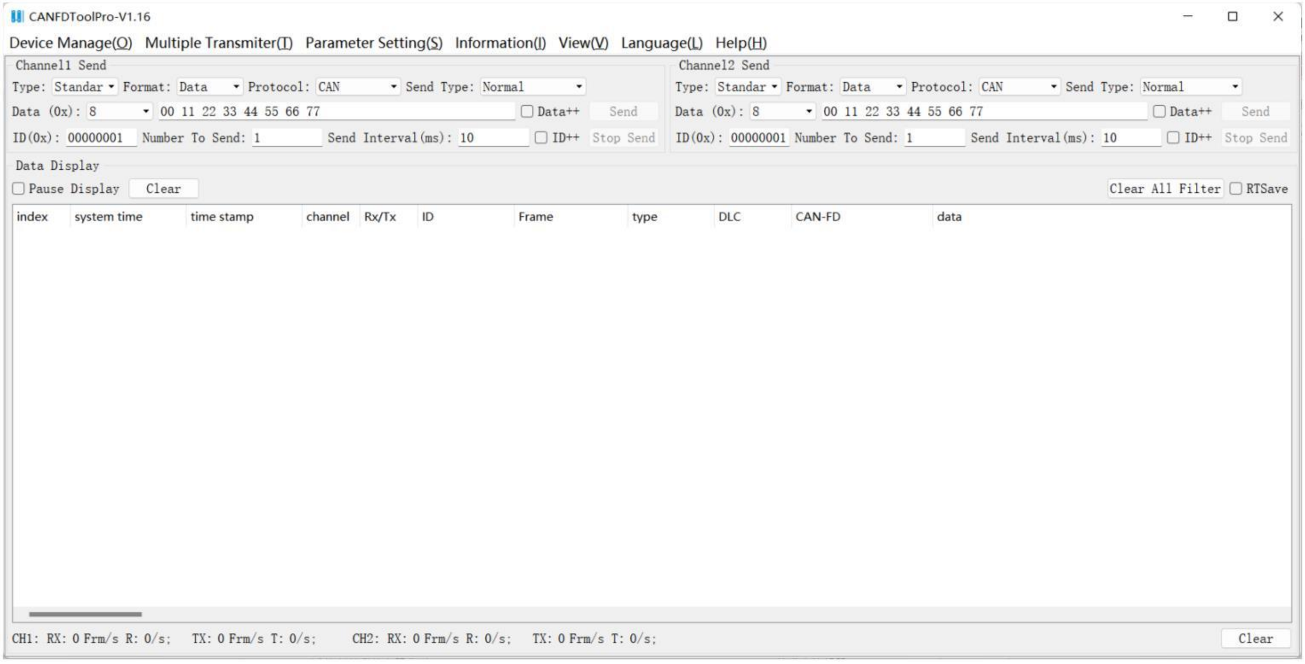
1、This device has a unique serial number that can be viewed through shell screen printing and CANFDToolPro software device information.
2、Optionally set the terminal resistance: by turning off the corresponding switch, the internal terminal resistance of 120 ohms can be used.
Note: Two 120 ohm terminal resistors must be ensured on the normal bus, otherwise it will affect the normal operation of the CAN bus.

3.2 Software Operation and Function Introduction

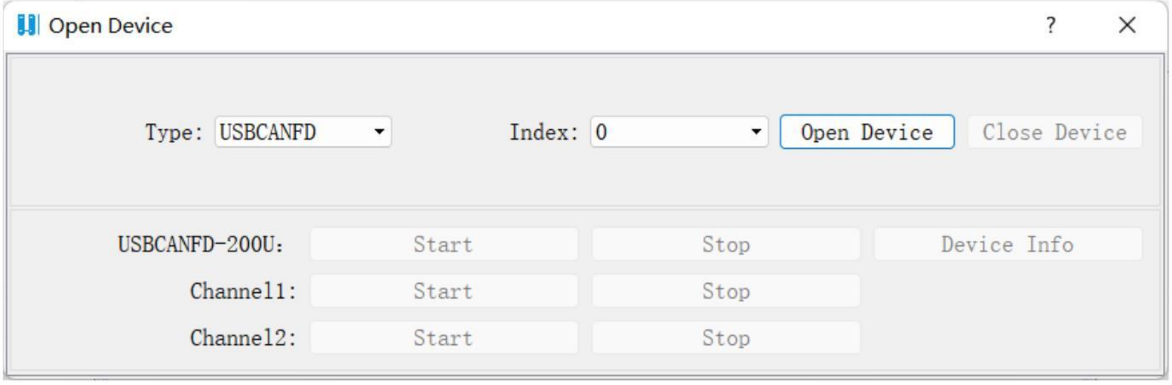
This device needs to be used in conjunction with the upper computer software (using the factory's built-in software CANFDToolPro.exe as an example).

Connect this device to the USB interface of the PC through a USB cable and run the tool software CANFDToolPro.exe, as shown in the following figure

Click on the "Device Management" ->"Device Operation" button to enter the device operation interface

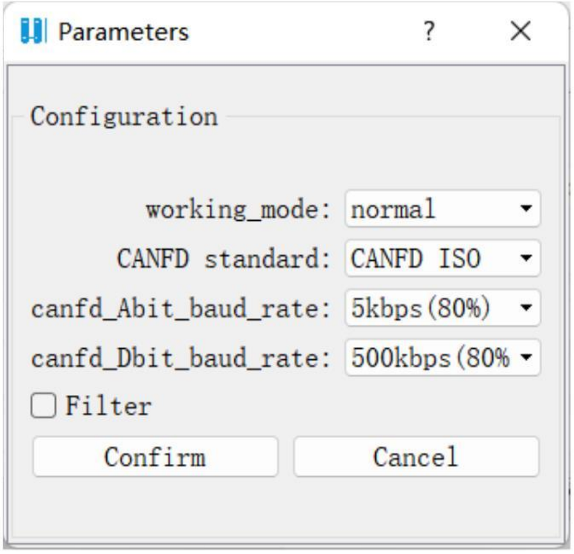


As shown in the figure, select the model and index number, click the "Open Device" button, and after successfully opening the device, you can configure the CAN channel. As shown in the following figure

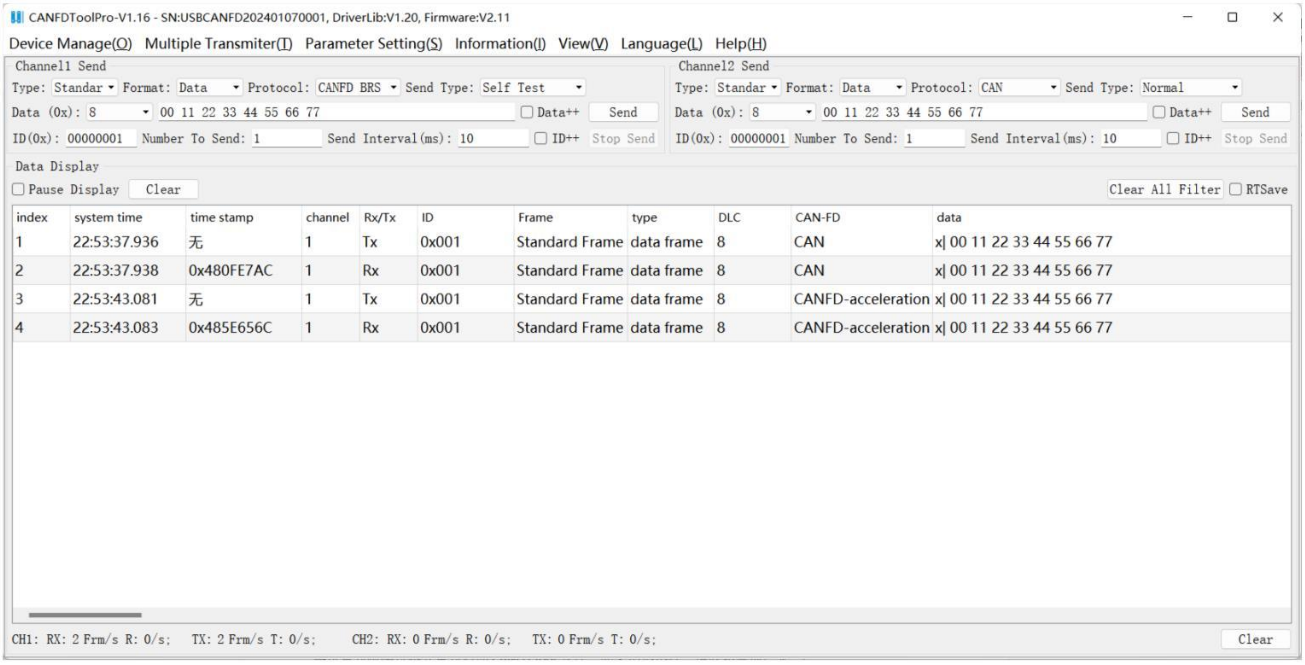


The three "start" buttons are: start all channels, start channel 1, and start channel 2. In the "Start" interface, configure the CAN parameters as shown in the following figure

After confirming the parameters, click the "Confirm" button to complete the settings.



You can select the transmission channel, transmission mode, frame format, protocol type, and data settings in the Home screen. Both successfully sent and successfully received data frames can be displayed in the Home screen. As shown in the following figure.



3.3 Transceiver testing

1、Connect the device to the USB interface of the PC and run the CANFDToolPro software.

2、Connect the CANH and CANL of the two channels of the device accordingly.

3、Open the device and start 2 channels according to Section 3.2 to send, and check whether there is data frame display in the Home screen. If both the sent and received data are displayed, it indicates that the device channel is normal.

Reminder:

If the above operation fails, please carefully check whether the wiring is correct, whether the matching resistance is enabled, and whether the Baud setting is correct. And you can try unplugging the device again and restarting the CANFD Tool software. If it still doesn't work, please contact the manufacturer for consultation.

3.4 About

This product is a key product launched by our company, with more practical and interesting features constantly being developed and updated. The newly released features will be promptly announced on the company's official website and Taobao store. If you need them, please contact technical support for updates.